```
import numpy as np

from mtx_utils import load_mtx

A, b = load_mtx("tasks/1_3.txt")
n = len(b)

A, b
```

```
(array([[ 29.,   8.,   9.,  -9.],
        [ -7., -25.,   0.,   9.],
        [  1.,   6.,  16.,  -2.],
        [ -7.,   4.,  -2.,  17.]]),
 array([ 197., -226.,  -95.,  -58.]))
```

Введём функции нормы

```
def matrix_norm(matrix):
    matrix = np.asarray(matrix)
    return np.abs(matrix).sum(axis=1).max()

def vector_norm(vector):
    vector = np.asarray(vector)
    return np.abs(vector).max()
```

Точное решение СЛАУ:

```
from IPython.display import display, Math

actual = np.linalg.solve(A, b)
for i in range(n):
    display(Math(fr"x_{i + 1} = {actual[i]:.3f}"))
```

$x_1 = 7.000$

$x_2 = 6.000$

$x_3 = -9.000$

$x_4 = -3.000$

Метод простых итераций

Найдём α и β

```
import pandas as pd

alpha = A.copy()
beta = b.copy()
```

```
for i in range(n):
    beta[i] /= alpha[i, i]
    alpha[i] /= -alpha[i, i]
    alpha[i, i] = 0

display(pd.DataFrame(alpha), pd.DataFrame(beta))
```

	0	1	2	3
0	0.000000	-0.275862	-0.310345	0.310345
1	-0.280000	0.000000	0.000000	0.360000
2	-0.062500	-0.375000	0.000000	0.125000
3	0.411765	-0.235294	0.117647	0.000000

	0
0	6.793103
1	9.040000
2	-5.937500
3	-3.411765

Зададим точность $\varepsilon = 0.001$

```
eps = 1e-3
```

```
actual = np.linalg.solve(A, b)
for i in range(n):
    display(Math(fr"x_{i+1} = {actual[i]:.3f}"))
```

$x_1 = 7.000$

$x_2 = 6.000$

$x_3 = -9.000$

$x_4 = -3.000$

```
norm_alpha = matrix_norm(A)
norm_beta = vector_norm(b)

iteration = 1
prev_x = beta.copy()
```

```
while True:
    x_iteration = beta + alpha @ prev_x
    eps_i = vector_norm(x_iteration - prev_x)
    if norm_alpha < 1:
        eps_i *= norm_alpha / (1 - norm_alpha)
```

```
    if eps_i <= eps:
        break

    prev_x = x_iteration
    iteration += 1
```

```
for i in range(n):
    display(Math(fr"x_{i+1} = {x_iteration[i]:.3f}"))

print(f"{iteration} итераций")
```

$x_1 = 7.000$

$x_2 = 6.000$

$x_3 = -9.000$

$x_4 = -3.000$

17 итераций

Метод Зейделя

```
C, B = np.triu(alpha), np.tril(alpha, k=-1)
# верхняя, с главной диагональю и нижняя, без главной диагонали
norm_c = matrix_norm(C)

iteration = 1
prev_x = beta.copy()
```

```
while True:
    x_seidel = prev_x.copy()

    for i in range(n):
        x_seidel[i] = beta[i] + alpha[i] @ x_seidel

    eps_i = vector_norm(x_seidel - prev_x)
    if norm_c < 1:
        eps_i *= norm_c / (1 - norm_c)

    if eps_i <= eps:
        break

    prev_x = x_seidel
    iteration += 1
```

```
for i in range(n):
    display(Math(fr"x_{i+1} = {x_seidel[i]:.3f}"))

print(f"{iteration} итераций")
```

$x_1 = 7.000$

$x_2 = 6.000$

$x_3 = -9.000$

$x_4 = -3.000$

```
9 итераций
```

Проверка решения

```
np.allclose(x_seidel, x_iteration, eps)
```

```
True
```

```
np.allclose(x_seidel, actual, eps)
```

```
True
```